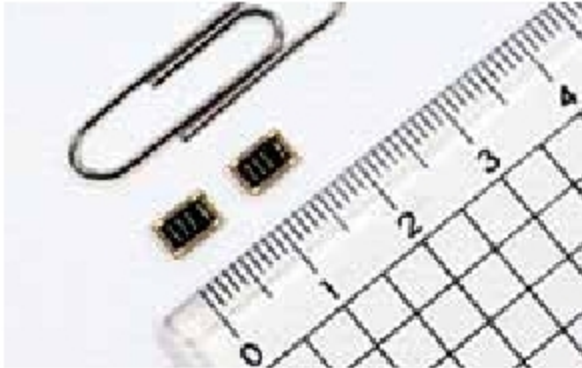


such as glucose measuring patches, smart lighting, outlets, switches, wireless earphones, and hearing aids. Using its low-power consumption feature, the module can be used for IoT devices that require a limited power supply, such as small wearable devices.



By linking various IoT devices to the smartphone, users can conveniently track blood glucose levels, remotely control lighting, outlets, and switches, and operate wireless earphones or hearing aids at a short distance, claims the company.

LG Innotek
www.lginnotek.com

Secure IC chip

With an increase in IoT devices across various fields, data attacks have consequently become rampant. In order to protect these devices, an authentication key is conventionally written in an IC memory. While this is an efficient way of securing data against any unauthorised breaches, attackers nowadays have become quite advanced and are using sophisticated malicious techniques to retrieve information.

To deal with such security issues, a multifunctional and secure IC has been



developed for equipping electronics designers to provide enhanced security measures for several IoT, IIoT, and medical devices.

Launched by Panasonic Corporation, the IC comes with near-field communication (NFC) for short-range wireless communication and resistive random-access memory (ReRAM) that is highly resistant to radiation.

Panasonic
www.panasonic.com

Automotive touch screen controllers

To help enhance and ease modern driving experience, touch displays play a major role beyond the centre infotainment display (CID). Supporting the application of these secondary displays with advanced features, Microchip Technology has now extended its maXTouch portfolio with the inclusion of the new MXT288UD touch controller family.



The MXT288UD-AM and MXT144UD-AM automotive-grade packaged touch screen controllers offer low power mode, weatherproof operation, and glove touch detection in multi-function displays, touchpads, and smart surfaces for vehicles, motorcycles, e-bikes, and car-sharing services.

The devices enable secondary touch surfaces to be placed in both the interior of cars and exterior of a motor vehicle, such as handlebars, doors, electronic mirrors, control knobs, the steering wheel, between seats, or in an armrest.

Microchip Technology
www.microchip.com

Miniature ceramic capacitor

The spread of 5G smartphones and the increasing miniaturisation of wearable devices have led to the demand for further miniaturisation and a higher density of electronic circuitry. Multilayer ceramic capacitors (MLCCs) are considered to be essential components in many kinds of electronic devices and are widely used.



Now, a new multilayer ceramic capacitor with a capacitance value of 1.0 μ F has been developed for a wide range of mobile electronic devices, including smartphones. These MLCCs enable a reduction of approximately 35 per cent in footprint and fifty per cent in volume ratio compared to existing products with the same capacitance value (266cm size).

Additionally, these have an increased capacity of about 2.1 times compared to conventional products of the same size (266cm size).

Murata
www.murata.com

All-in-one power management ICs

The recently launched all-in-one power management integrated circuits (PMICs) MUA01 and MUB01 offer a high-quality listening experience for today's True Wireless Stereo (TWS) devices.

The MUA01 and MUB01 PMICs have



been designed for the charging case and the earbuds, offering fully-integrated solutions optimised for TWS devices. To achieve this, up to ten discrete components were integrated into one, including switching chargers and discharge circuits, enabling the PMICs to occupy less than half the space compared to other alternatives.

This also allows compact solutions to allocate more room for the battery for longer playback time and enable more flexible and efficient designs.

Samsung Electronics
www.samsung.com

3-in-1 sensor for bezel-less displays

Here's a small integrated ambient light sensor (ALS) and proximity detection module that claims to enable mid-range mobile handset OEMs to develop mobile devices with virtually bezel-less displays.